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artist: A Python Package for AI-Enhanced Differentiable Raytracing in Solar Tower Power Plants

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Summary

artist is a Python software package for the simulation and optimization of concentrating solar power (CSP) plant operation. Solar tower power plants use an array of mirrors, known as heliostats, to reflect and concentrate sunlight onto a small receiver located at the top of a tower. The resulting thermal power, represented as a flux distribution on the receiver, can either be used directly as high-temperature heat in industrial processes or converted into carbon-neutral electricity. artist implements a fully differentiable digital twin of solar tower power plants and provides tools for data-driven power plant component modeling and heliostat aim point optimization. Its differentiable ray tracer simulates light transport within a three-dimensional scene in due consideration of environmental conditions, while enabling the integration of ray tracing into gradient-based optimization. The key functionality and workflow of artist are illustrated in Figure 1. In real-world operation, all physical plant components are subject to manufacturing tolerances and imperfections, which typically increase as the system ages. As a result, each heliostat produces an individually distorted flux distribution. To maximize plant efficiency, the real-world flux of every heliostat must be directed to a specific aim point on the receiver so that the combined flux density distribution becomes optimal. The main sources of uncertainty are small surface deformations (Ulmer et al., 2011) and misalignments in each heliostat caused by inaccuracies in the kinematic components (J. C. Sattler et al., 2020). These effects accumulate across the heliostat field and must be accounted for in accurate simulations. To address these challenges, artist provides functions to reconstruct real-world heliostat surface geometries and kinematic properties. It includes algorithms for alignment and ray tracing that enable the prediction of flux densities on the receiver. Building on this, heliostat aim points can be optimized to maximize the plant's overall energy yield.

Statement of Need

Digital twins of solar tower power plants with precise simulation and reliable predictive capabilities are essential for enabling fully autonomous operation and thereby reducing costs (Huang et al., 2021). While individual plants may differ in architectural details, their strong dependence on accurately modeling light paths and reflections makes ray tracing a key component. Conventional CSP ray tracers (Ahlbrink et al., 2012; Cardoso et al., 2018; Wendelin et al., 2018) achieve good results in simulating power plant behavior but are limited to making predictions through forward simulations based on the supplied data and underlying model. From a machine learning perspective, these ray tracers are thus restricted to forward computations and often require large amounts of data to achieve high accuracy. artist addresses this limitation through its differentiable implementation of the ray tracer and all connecting modules. Single differentiable digital twin tasks for CSP simulations have already been explored in related works, such as inverse deep learning approaches for heliostat surface modeling (Lewen et al., 2025) and generative neural networks for flux prediction (Kuhl et al., 2024). In contrast to these black-box AI methods, `artist` maintains interpretability through its foundation in physical models. Transparency in system behavior is critical for building trust in automated decision-making and software-generated operational recommendations. The methodology implemented in `artist` remains consistent across all optimization tasks, integrating modeling, prediction and optimization into a unified framework. `artist` is designed for researchers, power plant operators, developers within the CSP community, and anyone interested in the field. It provides data loaders compatible with various data sources, including the open-access CSP database PAINT (Phipps et al., 2025), enabling contributions from researchers who do not have direct access to an operational power plant. By developing `artist` as an easily accessible open-source software package, we aim to strengthen community engagement and foster collaboration for further research advancements.

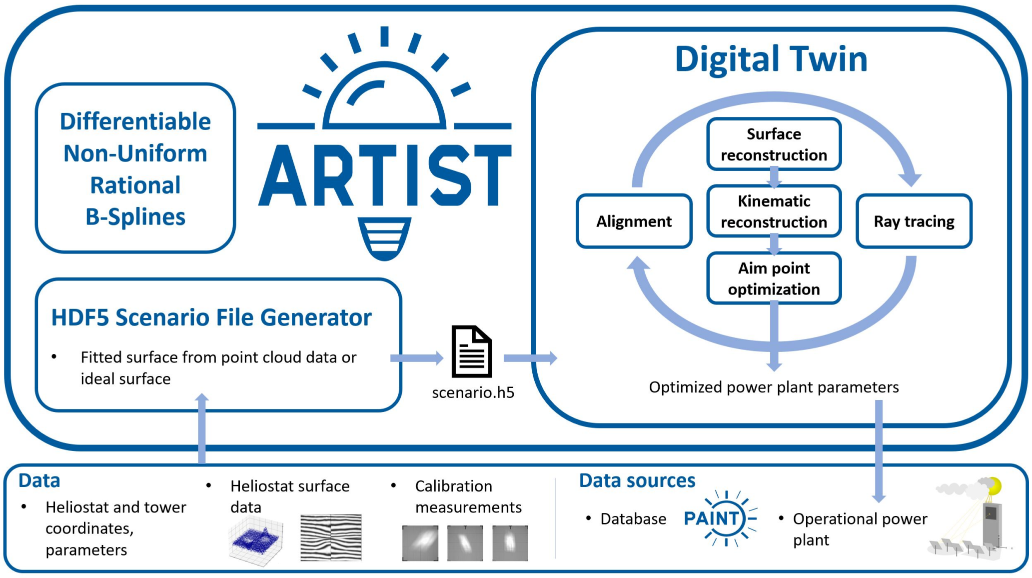

Figure 1: Features of `artist`, the **A**I-Enhanced Differentiable **R**ay **T**racer for **I**rradiation Prediction in **S**olar Tower Digital **T**wins (ARTIST). To create digital twins of solar tower power plants in `artist`, users provide HDF5 files containing data about the power plant's physical layout. These HDF5 scenarios can be generated by `artist` from various data sources. `artist` unpacks the files to initiate the reconstruction, prediction, and optimization. The optimized power plant parameters, i.e., the motor positions for each heliostat, can be used directly as input for power plant control software. To efficiently represent heliostat surfaces in the digital twin, `artist` includes a fully differentiable, parallelized NURBS implementation for three-dimensional surfaces.

State of the Field

Commercial digital twin software solutions for solar tower power plants, such as the raytracer STRAL (Ahlbrink et al., 2012) or TieSOL (Mitchell et al., 2025), are non-differentiable and therefore lack the ability to reconstruct heliostat field models or to optimize power plant parameters. Instead, they assume idealized hardware or provide tools for exact modeling which rely on practically infeasible measurements. Moreover, these commercially proven software solutions are typically proprietary and closed-source, limiting opportunities for public contributions. Even for open-source, commercially validated, non-differentiable ray tracers such as SolTrace (Wendelin et al., 2018), the modifications required to propagate gradients through the entire simulation would be so extensive and fundamental that the development of a new tool is justified. In literature, many proof-of-concept studies address single tasks relevant for digital twin simulation and optimization of solar tower power plants. These include differentiable ray tracing (Pargmann et al., 2024), surface modeling (Lewen et al., 2025), kinematic calibration (J. Sattler et al., 2024), and flux prediction (Kuhl et al., 2024). While each of these approaches demonstrates promising capabilities, they mostly remain isolated solutions. Since each task relies on a methodology specialized to its specific requirements, straightforward unification is challenging. `artist` addresses this gap by applying one coherent optimization strategy across all tasks, redefining and extending existing approaches to form an integrated and practicably usable software product that is fundamentally different from the task-specific solutions but solves the same problems. As no comparable open-source alternative exists, it provides a solid foundation for future contributions.

Software Design

`artist` is designed for computational efficiency by balancing memory consumption, execution time, and simulation accuracy. To preserve real-time capabilities and maintain scalable runtimes for power plants with several thousand mirrors, `artist` features native GPU acceleration and supports distributed computation. On a single GPU, `artist` parallelizes the computation of individual heliostats by storing data in the Structure of Arrays (SoA) format, ensuring contiguous memory layout. Combined with the GPU's Single-Instruction, Multiple-Thread (SMIT) processing model, this data structure enables coalesced memory access, minimizing memory overhead while maximizing GPU utilization. For distributed execution across multiple compute nodes, `artist` uses a nested data-parallel approach that distributes groups of heliostats among the available compute resources. Each distributed group can additionally be parallelized through nested sub-processes. Heliostats within a single group share mathematically identical alignment calculations that differ from those of other groups, which requires them to be processed separately for efficient parallelization. In `artist`, parameter learning relies purely on gradient-descent based optimization combined with physics-informed models and constraints. This combination stabilizes the optimization of the underdetermined system and reduces the search space. It also enables fast convergence with fewer samples, thereby reducing latency and memory usage. Users can configure key simulation and optimization parameters to adjust the trade-off between memory usage, computational time, and precision according to their needs. The core algorithms in `artist` are implemented using fully differentiable methods building on PyTorch's automatic differentiation system for gradient propagation. Overall, `artist` adheres to the FAIR principles for research software (FAIR4RS) (Barker et al., 2022) and ensures portability across multiple hardware platforms by leveraging CI/CD pipelines with automated tests on Windows, Linux, and MacOS. Even though `artist` is optimized for GPU execution, it supports both CPU and GPU operation and automatically selects the appropriate communication layer for multi-node parallel execution based on the underlying operating system and hardware configuration. Its modular architecture, built on abstraction and inheritance, enables application across diverse solar tower power plant designs. Users can extend existing interfaces to incorporate plant-specific design details into heliostat field models. The `artist` software package is fully documented via Read the Docs, including installation instructions, tutorials, theoretical background on mathematical concepts and data structures as well as a complete API reference.

Research Impact Statement

The underlying concepts of `artist` build on previous publications that demonstrate the potential of the differentiable ray tracing approach (Pargmann et al., 2024). More generally, these works highlight the potential for improving solar tower power plant efficiency as a means of providing an environmentally friendly solution to meet the globally rising demand for energy (Carballo et al., 2025; Edenhofer et al., 2011; Zhu et al., 2023).

AI Usage Disclosure

Generative AI was employed solely as an editorial and technical aid, specifically for debugging code and refining the manuscript. Generative AI did not contribute to the underlying architectural design or any scientific findings.

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